

The perimeter of a semicircle and of a quarter circle

Half the curve is not the whole perimeter — the straight edges have to be added.

LESSON 95–96

2 × 40 MIN

MATEMATIKA 6, §20

S7 15

LESSON CLOCK

20'

25'

25'

8'

2

Arc length	20 min
The semicircle	25 min
The quadrant	25 min
Compound shapes	8 min
Homework	2 min

LEARNING OBJECTIVES

- Find the length of an arc as a fraction of the circumference.
- Find the perimeter of a semicircle and of a quadrant.
- Find the perimeter of a shape made of arcs and straight edges.
- Distinguish arc length from perimeter.

TEXTBOOK

National	pp. 273–280
Cambridge	Stage 7, pp. 148–154
Workbook	Exercise 15.1

01

Explanation

Arc length

arc = $\frac{\alpha}{360} \times C$

CENTRAL ANGLE	FRACTION	ARC OF A CIRCLE WITH $C = 44$
180°	$\frac{1}{2}$	22
90°	$\frac{1}{4}$	11
120°	$\frac{1}{3}$	14.67
45°	$\frac{1}{8}$	5.5

EVERYTHING IS A FRACTION OF THE WHOLE CIRCLE

The fraction is the central angle over 360. Nothing else about arcs is needed at this stage.

The semicircle

The perimeter of a half disc is the curved half **plus the diameter**.

$P = \pi r + 2r$

R	CURVE πR	DIAMETER	PERIMETER
7	22	14	36
14	44	28	72
10	31.4	20	51.4

HALF THE CIRCUMFERENCE IS NOT THE PERIMETER

A half disc is bounded by a curve *and* a straight edge. Leaving out the diameter is the commonest error in the whole topic — and the shape it describes would be open.

The quadrant

A quarter disc is bounded by a quarter of the curve and **two radii**.

$P = \frac{1}{2}\pi r + 2r$

R	CURVE	TWO RADII	PERIMETER
7	11	14	25
14	22	28	50
10	15.7	20	35.7

COUNT THE STRAIGHT EDGES FROM THE PICTURE

A semicircle has one; a quadrant has two. Drawing the shape and tracing round it with a finger is a more reliable method than remembering formulae.

Compound shapes

SHAPE	PIECES OF THE BOUNDARY	PERIMETER WITH $R = 7$
a semicircle on a 14 by 10 rectangle	three sides plus the curve	$10 + 10 + 14 + 22 = 56$
a running track: two straights of 80 and two semicircular ends of radius 35	two straights plus a full circle	$160 + 220 = 380$
a quadrant cut from a square of side 7	two sides plus the curve	$7 + 7 + 11 = 25$

ONLY THE OUTSIDE COUNTS

Where the semicircle sits on the rectangle, the 14 cm edge is inside the shape and is not part of the perimeter — but the two ends of it are. Trace the boundary before adding anything.

02 Worked examples

EXAMPLE 1 Find the perimeter of a semicircle of radius 7 cm, taking $\pi = \frac{22}{7}$.

1

The curve: $\pi r = \frac{22}{7} \cdot 7 = 22$ cm.

2

The straight edge is the diameter: 14 cm.

3

$22 + 14 = 36$ cm.

ANSWER

36 cm

EXAMPLE 2 Find the perimeter of a quadrant of radius 14 cm.

① The circumference is $2 \cdot \frac{22}{7} \cdot 14 = 88$ cm.

② A quarter of it: 22 cm.

③ Two radii: 28 cm.

④ $22 + 28 = 50$ cm.

ANSWER*50 cm***EXAMPLE 3 A running track has two straights of 80 m and two semicircular ends of radius 35 m. Find one lap.**

① The two semicircles make one full circle.

② $2 \cdot \frac{22}{7} \cdot 35 = 220$ m.

③ The straights: 160 m.

④ $220 + 160 = 380$ m.

ANSWER*380 m***03 Interactive model**

Have the class trace the boundary of each shape with a finger and say aloud what they pass; the missing diameter stops being missed.

Curve plus straight edges

A semicircle of radius 7 has curve of length:

Its perimeter is:

A quadrant of radius 7 has curve of length:

Its perimeter is:

11

18

25

32

A semicircle of radius 14 has perimeter:

44

72

88

28

A 120° arc of a circle with $C = 44$ is:

11

14.67

22

5.5

Two semicircular ends of a track make:

half a circle

one full circle

two circles

a quadrant

The commonest error here is:

using π

forgetting the straight edges

using the radius

adding twice

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Perimeter	Perimetr	Периметр
Arc length	Yoy uzunligi	Длина дуги
Semicircle	Yarim doira	Полукруг
Quadrant	Chorak doira	Четверть круга
Straight edge	To'g'ri tomon	Прямая сторона
Compound shape	Murakkab shakl	Составная фигура
Fraction of the circle	Aylananing qismi	Часть окружности
To add the edges	Tomonlarni qo'shish	Сложить стороны

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

An arc of 90° is what fraction of the circle?

$$\frac{1}{2}$$

$$\frac{1}{3}$$

$$\frac{1}{4}$$

$$\frac{1}{8}$$

The perimeter of a semicircle is:

$$\pi r$$

$$\pi r + 2r$$

$$2\pi r$$

$$\pi r + r$$

The perimeter of a quadrant is:

$$\frac{1}{2}\pi r$$

$$\frac{1}{2}\pi r + 2r$$

$$\frac{1}{4}\pi r + r$$

$$\pi r + 2r$$

A semicircle of radius 10 has perimeter about:

31.4

51.4

62.8

20

Two semicircular ends of equal radius make:

a semicircle

a full circle

a quadrant

nothing

A boundary edge inside a compound shape:

- counts
- does not count
- counts twice
- is halved

06

Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. A semicircle of radius 7: the curve

ANSWER 22

2. Its perimeter

ANSWER 36

3. A semicircle of radius 14: the curve

ANSWER 44

4. Its perimeter

ANSWER 72

5. A quadrant of radius 7: the curve

ANSWER 11

6. Its perimeter

ANSWER 25

7. A 90° arc as a fraction of the circle

ANSWER $\frac{1}{4}$

Medium · the standard question

7 PROBLEMS

1. A quadrant of radius 14: its perimeter

ANSWER 50

2. A semicircle of radius 10 ($\pi = 3.14$)

ANSWER 51.4

3. A quadrant of radius 10

ANSWER 35.7

4. A 120° arc of a circle with $C = 44$

ANSWER 14.67

5. A 45° arc of the same circle

ANSWER 5.5

6. A track: straights 80 m, ends of radius 35 m

ANSWER 380 m

7. A quadrant cut from a square of side 7

ANSWER 25

Hard · stretch and reason

7 PROBLEMS

1. A semicircle on top of a 14 by 10 rectangle: the perimeter

ANSWER 56

2. A shape made of three quadrants of radius 7

ANSWER Curve 33 plus the straight edges

3. A semicircle of perimeter 36: its radius

ANSWER 7

4. A semicircular window of diameter 1.4 m: the frame length

ANSWER 3.6 m

5. A running track of two straights 100 m and ends of radius 28 m

ANSWER 376 m

6. Why add the diameter to a semicircle?

ANSWER The straight edge is part of the boundary

7. A quarter-circle path of radius 21 m: its curved length

ANSWER 33 m

07 Homework

Homework — 5 tasks

Trace the boundary with a finger and list every piece before adding.

1. Find the perimeter of a semicircle of radius 21 cm.
2. Find the perimeter of a quadrant of radius 21 cm.
3. Find the length of a 60° arc of a circle of radius 7 cm.
4. A semicircle sits on top of a rectangle 20 cm by 12 cm, on the 20 cm side. Find the whole perimeter.

5. A track has straights of 90 m and semicircular ends of radius 21 m. Find one lap.